EXPERIMENT - 03

Aim: To study CRO and function generator with specifications.

Apparatus: CRO, function generator, connecting wires, Resistors capacitors , diode.

Theory: The cathode Ray ascilloscope [CRO] is an electronic instrument used to observe and measure electrical signals such as voltage frequancy, and wave form shape, it displays signals on a screen in real time, helping analyze how signals charge over time. The CRO works by emitting an electron beam from a heated cathod, which is accelerated and directed onto a fluorescent screen. imp parts include the Y-input Lvertical, vertical attenvator, vertical amplifier and delay line which together control and display the signal clearly on the screen.

Procedure:

(1) Set the value of the function generate 0 frequency which is connects to CRO.

(2) Net in the line wave length ladjust a laud look in the CRO.

(3) Note values of frequency and time is i/p function generator.

(4) note the reading of the output from CRO of amplitude and frequency.

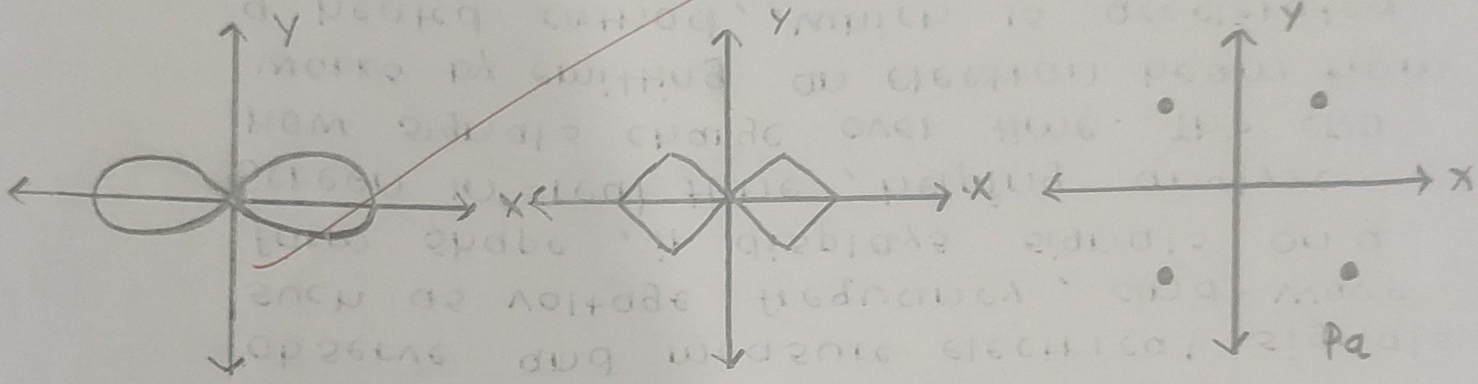
(5) Now set in square and measure sample input.

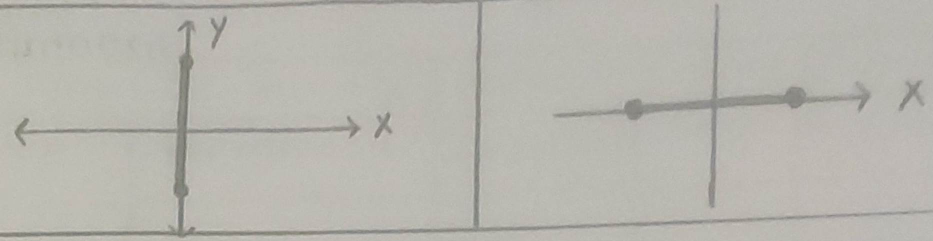
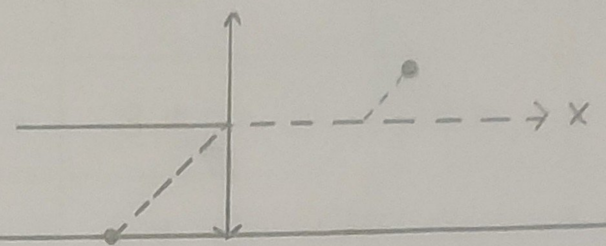
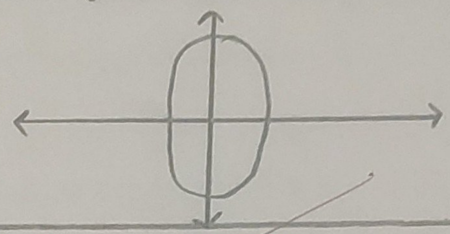
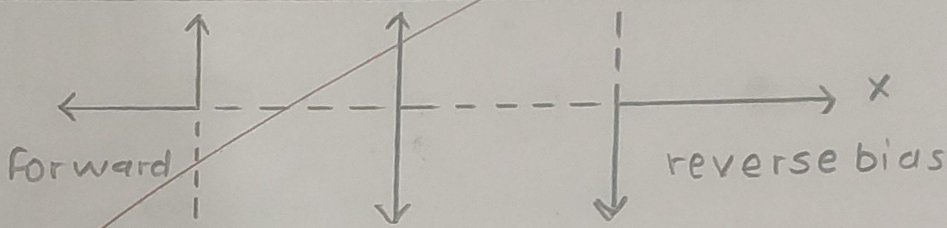
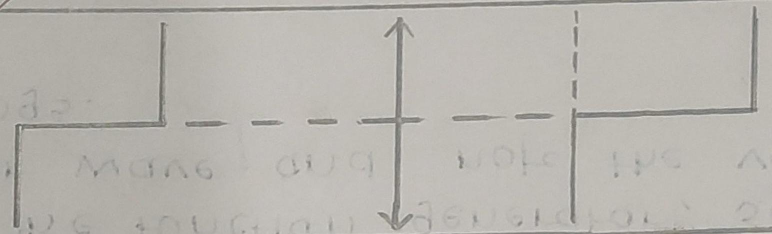
(6) set the function generator, source to trigger wave and note the various readings.

Observation Tabel:

input signal	Input value [from fun generator]			measured value [from CRO]		
	Amplitude	Time	Frequency	Amplitude	Time	Frequency
sine wave	10 V	1 ms	1 KHZ	10 V	1 ms	1 KHz
square wave	15 V	0.1 ms	3 KHz	15 V	0.1 ms	3.1 KHz
Triangle wave	8 V	0.5 ms	0.7 KHz	8 V	0.5 ms	0.714 Hz

- To observe and draw waveform of Lissajous pattern:



(1) short circuit (2) open circuit	
Resistor	
capaciton	
Diode	
zener diode	

Calculation: (1) sine function : $T_1 = 2 \times 0.5 \times 10^{-3}$ m/s

$F_1 = 1$ KHz

(2) square $T_2 = 3.2 \times 0.1 \times 10^{-3}$

$f_2 = \dfrac{1}{3.2 \times 10^{-3} \times 0.1} = 3.12$ kHz

(3) Trianglar

$T_3 = 2.8 \times 0.5 \times 10^{-3}$

$F_3 = 7.14$ kHz

(4) Amplitude : total no of blocks x volt / div in y axis

$4 \times 0.2 = 0.8$ V

(5) Time : total no. of blocks x Time / div in a x-axis

$2 \times 0.5 = 1$ ms

Conclusion :

from this experiment learn that to measure voltage time period CRO the working process of CRO and instrument about function generator.